College students' willingness to participate in "time bank" voluntary service and its influencing factors

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***Abstract*—As an alternative economic model, the "time bank" aims to alleviate social problems through time money. Given that college students have more time and higher participation potential, they are crucial time bank participants. This paper aims to study the willingness of college students to participate in the time bank model of voluntary service and systematically explores the key factors affecting willingness to participate. This paper uses factor analysis and binary logistic regression analysis to comprehensively analyze the influencing factors from the questionnaire survey data of 628 college students' willingness to participate in the time bank from four levels: individual, family, social, and service and technology. The results revealed that the four levels all have significant positive effects on the willingness to participate in time banking. Specific factors at each level, such as grade, political status, whether the volunteer service experience is pleasant, freshman status, health, volunteer experience, and free days in the semester, had a significant positive impact on willingness, and the number of immediate elderly family members had a significant negative impact on the willingness. In contrast, while gender, monthly family income, and understanding of time bank have theoretical potential, they did not have significant effects in this study. This study provides a basis for policy-makers and managers to formulate more effective incentive mechanisms and participation strategies as well as suggestions for the development and optimization of time banking mechanisms in the future. In addition, this paper provides a theoretical basis for improving community services to enhance community cohesion and social integration.**

Keywords-Volunteer service; Time bank; Willingness to participate

I. INTRODUCTION

"Time banking" is an alternative economy in which the time-based exchange of monetary stimulus solves social problems and alleviates social problems by building up social capital [1]. Time banking was proposed in 1980 by Edgar Cahn, who proposed "time money" as an alternative to money to measure the value of people's service labor as a currency that could be saved and circulated. In recent years, with China's aging population and the popularity of volunteerism, time banks have attracted renewed interest.

The rewarding and reciprocal nature of time banking and the altruism and voluntarism of traditional volunteering are complementary [2]. In time banking, the "pay" is reflected in the social recognition and spiritual rewards for helping others. Even if material returns are involved,these also carry special social value. Two-way mutual assistance is foundational to time banking. In this system, each participant can save his volunteer service time to form "time currency," which can be exchanged for other volunteers, community services, or goods in the form of "points exchange" when needed or for various services (free haircuts, medical examinations, food discounts, etc.,) [3]. It is not limited to a specific age group but encourages people of all ages in society to participate in volunteering activities. This model breaks through the limitation of the traditional one-way volunteer service and forms a relaying chain of mutual assistance. It is a novel practice combining volunteer service and an incentive mechanism, which promotes the efficient use of resources and community cohesion [4].

The Opinions of The General Office of the State Council on Promoting the Development of Elderly Care Services issued by

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The General Office of the State Council in 2019 stated that efforts should be made to cultivate a team of elderly care volunteers, explore specific methods of time banking, and conduct time banking work in many governments. Wuhan City has explored the point model of a time bank, converting volunteer service hours into points and depositing them in the time bank. The points in the time bank can be exchanged not only for daily necessities, such as toothpaste and paper towels, but also for life services such as haircuts and home appliance repairs, creating a social culture of "repay virtue with virtue" In 2022, Shandong Women's University decided to implement the time bank concept and jointly launched Time Bank, a public volunteering platform, with the Bank of China. Through resource exchange, students can earn corresponding points by accumulating volunteer labor, which can be exchanged for credits for labor courses, innovation and entrepreneurship, and some prizes. Moreover, the time bank enhances students' sense of obligation and identity, promotes communication among them, builds an exchange and sharing platform, and improves students' practical skills.

In the time bank policies issued by Beijing, Shanghai, and other places, young people over 18 years old and in good health are encouraged to participate in the time bank service. College students, as the main reserve force behind the time bank volunteer team, can play an important role in volunteer service due to their flexible learning time, rich knowledge, and age advantages [5].

Therefore, this paper focuses on college students' willingness to participate in the time bank volunteer service and identifies the influencing factors to provide a new perspective on college students' motivation for participation in the time bank volunteer service. The results serve as a basis for policy-makers and managers to formulate more effective incentive mechanisms and participation strategies, sensitize college students regarding its importance for personal growth, and attract more young people to participate in volunteer service [6]. In addition, this paper will also provide a theoretical basis for improving community service to enhance community cohesion and social integration.

II. Literature Review

In the research of social participation and civic activities, Verba et al. proposed the Civic Volunteerism Model in 1995, which aimed to explore the impact of socioeconomic status on citizens' engagement in public affairs. Subsequently, some scholars introduced this model into their studies on the factors influencing willingness to participate in time banking and concluded that individual differences, family capabilities, health conditions, and policy environments were all potential influencing factors. Following this conclusion, this study divides the research on time banking into four dimensions: personal level, family and demographic level, social level, and service and technology level [7].Based on the research sample, this study hereby proposes the hypothesis:

H1: Individual, family, social, and service and technology level and other dimensions are positively correlated with college students' willingness to participate in the time bank mode of volunteer service.

A. Personal Level

In terms of personal factors, an individual's health status plays a crucial role in their willingness to engage in time banking. Research indicates that volunteers' willingness to continue volunteering increased significantly (by 0.132, $P<0.1$) with improvements in their fitness level [8]. Research based personal level indicates that the willingness to participate in time banking is significantly influenced by a combination of utilitarian, hedonic [9], and social values, alongside the pursuit of altruism and self-actualization [10]. Studies also have shown that altruistic motivations are particularly strong among college students, who are driven by the social contributions of time banking [11]. The social value of time banking notably impacts service providers, whereas the utilitarian value predominantly affects service recipients. When individuals perceive tangible benefits, such as acquiring practical knowledge and skills through time banking, their motivation to participate is notably heightened [12]. Young groups, such as students, are especially sensitive to intrinsic rewards, and their willingness to engage increases when they recognize personal gains and satisfaction from their involvement [13]. Furthermore, an individual's cognitive appraisal of time banking and prior volunteering experiences directly affect their willingness to participate, with previous volunteering experience positively influencing youth participation in online time banking [14]. Despite this, low participation rates persist due to a lack of awareness about the operational mechanisms and benefits of time banking [15]. Establishing trust and clearly communicating the benefits are crucial to enhancing participation, especially to new volunteers. Participation can also be hindered by limited initiative and creativity [16]. Allowing individuals to independently choose how to distribute the fruits of their service labor can improve their service experience and subsequently enhance their willingness to participate in time banking [17]. Research also demonstrated that students from middle economic backgrounds often have more leisure time, which positively influences their engagement in time banking activities [18]. The sustained operation of time banking systems helps establish a strong foundation and improves public willingness to participate.

In summary, the analysis of individual participation in time banks highlights the multifaceted and complex nature of personal motivations. Health status is a fundamental prerequisite, with better physical condition promoting greater involvement. The interplay between utilitarian, hedonistic, and social values, combined with altruistic and self-fulfillment drives, significantly shapes individuals' willingness to contribute to time banks. Notably, altruistic motives are prominent among university students, reflecting a generational trend towards socially responsible behavior. Perceived tangible benefits, such as skill acquisition and personal satisfaction, serve as strong motivators, particularly among young people who respond positively to intrinsic rewards. Therefore, this paper hypothesizes the following:

H2a: Time bank knowledge positively correlates with college students' willingness to participate in volunteer services under the time bank model.

H2b: Health status positively correlates with college students' willingness to participate in volunteer services under the time bank model.

H2c: Political status (e.g., involvement in student organizations, leadership roles) positively correlates with college students' willingness to participate in volunteer services under the time bank model.

H2d: Prior voluntary experiences positively correlate with college students' willingness to participate in volunteer services under the time bank model.

H2e: Being a new student positively correlates with college students' willingness to participate in volunteer services under the time bank model.

H2f: Number of free days during the semester positively correlates with college students' willingness to participate in volunteer services under the time bank model.

B. Family and Demographic Level

Family factors and demographic characteristics, such as age and gender, significantly influence the willingness to participate in time banking [19]. Research indicates a strong positive correlation between the number of immediate elderly family members and young people's willingness to engage in online time banking, suggesting that family structure enhances the younger generation's propensity to volunteer. Furthermore, gender differences in willingness to participate vary across age groups [20]. While older men are generally more willing to participate than women, this gender difference diminishes among younger individuals [21]. Among college students, women display a higher willingness to participate compared to men, likely due to their greater cognitive advantages in patience and empathy [22].

Age also plays a crucial role, exhibiting a hierarchical effect on participation willingness [23]. For the elderly, there is a notable decrease in willingness with increasing age, whereas, in younger age groups, age does not significantly impact participation intentions. Additionally, freshmen in college, who often have more free time and enthusiasm, are more likely to engage in time banking activities [24]. The cultural context of traditional Chinese values, emphasizing the connection between family and country, further reinforces the importance of family support in driving participation [25]. It highlights how familial encouragement and support can be pivotal in motivating individuals to participate in time banking.

In summary, family factors and demographic characteristics, including age, gender, and family structure, emerge as significant determinants of individuals' willingness to participate in time banking. The presence of elderly family members fosters a sense of responsibility and prompts younger generations to engage in volunteering, underscoring the role of family dynamics in shaping volunteering behavior. Age and gender disparities in participation intentions reflect complex generational and societal trends, with older individuals and younger women exhibiting heightened willingness in certain contexts. Given that insight, it is plausible to hypothesize that factors related to family demographic factors.

H3a: The number of immediate elderly family members positively correlates with college students' willingness to participate in time banking volunteer services.

H3b: Among college students, women exhibit a higher willingness to participate in time banking volunteer services compared to men.

C. Social Level

Social and economic conditions significantly affect individuals' willingness to participate in volunteer service [26]. Social support, including recognition and encouragement from educational institutions and peers, plays a crucial role in motivating volunteering [27]. Additionally, endorsements from celebrities and favorable public opinion, amplified through social media, substantially enhance young people's interest in time banking [28].

Despite these factors, time banks face challenges due to insufficient central government support and legal protection, which negatively impacts participation levels [29]. Effective government supervision and regulation could bolster the reputation of time banks, improve legal frameworks, and reinforce trust mechanisms, ultimately increasing public willingness to participate.

Based on the comprehensive review of the literature, it is evident that social and economic factors, along with social support and recognition, play pivotal roles in shaping individuals' willingness to engage in volunteer service through time banking systems. While students from middle economic backgrounds benefit from increased leisure time that fosters their participation, the sustained operation and strong foundation of time banking systems further enhance public interest and willingness to contribute. Thus, this study hereby propose hypothesis.

H4: Positive social support and guarantees, positively correlates with college students' willingness to participate in volunteer service through the time banking model.

D. Service and Technical Level

Time banks leverage long-term incentive mechanisms to cater to diverse individual expectations and enhance participant engagement [30]. However, the principle of time equivalence—measuring all labor equally—can oversimplify value assessment, potentially leading to perceived unfairness and reduced participation [31]. Geographic limitations and restricted time money circulation further constrain the appeal of time banks, which can be mitigated by offering accessible life services such as cleaning and shopping, thus broadening participation. Utilizing Internet technology, time banks can create decentralized, interactive platforms that improve supply and demand matching, enhance volunteers' self-efficacy, and protect time money circulation, thereby sustaining effective incentive mechanisms and boosting participant willingness [32]. Accordingly, this study sets a hypothesis related to the experience at the service and technology levels, that it is

Identify applicable sponsor/s here. *(sponsors)*

H5: The satisfaction derived from volunteer experiences positively correlates with college students' willingness to engage in the volunteer time bank model.

Except comprehensive research on time bank participation in elderly care, significant gaps remain. Literature predominantly focuses on elderly services, with limited exploration into broader voluntary applications like environmental protection, education, and community building. Expanding research into these areas could reveal the model's diverse potential and challenges across various social needs, such as environmental projects and educational equity. Additionally, current studies often overlook deeper factors like experience value, which could influence college students' decision-making and participation. While existing research provides detailed descriptive analysis, it lacks exploration of underlying causal relationships and specific influencing mechanisms.

III. Research Design and Data Analysis

A. Sample Selection

In this study, the research team conducted a questionnaire survey on the willingness of Chinese college students to participate in time banking. The criteria for selecting the survey participants were Chinese college students willing to participate and fluency in communication. This study collected 628 data and 628 valid questionnaires, as listed in Table I. The 628 Chinese college students surveyed had a balanced gender distribution, with a similar proportion of males and females. Freshmen accounted for 67.5% of the respondents, 66.2% were student leaders, and the average number of free days per week this semester was 2.7. Moreover, 65% of the participants were willing to continue participating in the volunteer service and volunteer service experience, and most of the monthly family income was between 5,000 and 15,000.

B. Variable Description

Table I. Basic Information Frequency Statistics Table

Names	Sample	Minimum	Maximum	Average	Standard Deviation	Median
Gender	628	0	1	0.56	0.50	1
Freshmen	628	0	1	0.68	0.47	1
Student cadre	628	0	1	0.66	0.47	1
Political profile	628	1	5	2.59	1.41	2
Good physical health or not	628	0	1	0.64	0.48	1
Volunteer experience	628	0	1	0.66	0.47	1
Free days a week during the semester	628	1	4	2.76	0.95	3
Volunteering experience is enjoyable or not	628	0	1	0.65	0.48	1
Be willing to participate in volunteering on an ongoing basis or not	628	0	1	0.66	0.48	1
Community Assistance Services	628	0	1	0.55	0.50	1
Supporting education in poverty alleviation	628	0	1	0.25	0.43	0
School admissions publicity	628	0	1	0.52	0.50	1
Environmental activities	628	0	1	0.36	0.48	0
Professionally related volunteering	628	0	1	0.35	0.48	0
Charity fundraisers	628	0	1	0.43	0.50	0
Volunteering on campus	628	0	1	0.31	0.46	0
Volunteer activities for the elderly	628	0	1	0.32	0.47	0
Monthly household income level	628	1	4	2.54	0.79	3
Number of immediate elder families	628	1	5	2.74	1.21	3
Knowledge level of time bank	628	1	5	3.63	1.36	4
Willingness	628	0	1	0.71	0.45	1
Learning resources	628	0	1	0.35	0.48	0
Career development opportunities	628	0	1	0.38	0.49	0
Skills development	628	0	1	0.26	0.44	0
Service	628	0	1	0.32	0.47	0
Change	628	0	1	0.33	0.47	0
Supplies	628	0	1	0.25	0.43	0
Credits	628	0	1	0.50	0.50	0.5
Other	628	0	1	0.09	0.28	0

The descriptive statistical data of the important dimensions of Chinese college students' willingness to participate in volunteer service in the time bank model are listed in Table II. In this study, the dependent variable is “willingness to participate in time banking.” The independent variable comprises the following four dimensions. First, the personal level refers to the personal factors that affect college students’ willingness to participate in volunteer service in the time bank model. The research has revealed that personal values, health status, volunteering experience, and knowledge of time banking affect college students’ willingness to participate in time banking [33]. Family level measures the influence of factors such as family members’ support, family structure, gender, and age on college students’ willingness to participate in time banking [34]. The social level is closely related to college students’ willingness to participate in time banking, who are the base of time banking, based on social and economic conditions, social support networks, public opinion, strong government support, and legal guarantees [35]. Finally, at the service and technology level, the mechanism and technological innovation behind the time bank have an important effect on their participation willingness. Research

has shown that the operation mechanism of time banking, the return mechanism, the reduction of geographical restrictions, and the use of Internet technology to optimize the service process can effectively improve the participation willingness of participants [36].

Through factor analysis and binary logistic regression analysis of the questionnaire survey data, this paper highlights the factors that play a key role in promoting or hindering Chinese college students' participation in time banking to provide a theoretical basis for relevant policy formulation and community service improvement.

Table II. MULTI-DIMENSIONAL STATISTICAL TABLE

Dimensions	Name	Sample size	Minimum	Maximum	Average	Standard Deviation	Median
Individual level	Receive a stipend from the sponsor or school	628	0	5	1.416	1.271	□
	Get the corresponding recognition from the sponsor or the school	628	0	5	1.36	1.252	□
	Get credit	628	0	5	1.403	1.252	□
	Opportunities to explore different careers and add to my resume	628	0	5	1.366	1.242	□
	Broaden horizon, exercise ability and increase social experience	628	0	5	1.439	1.318	□
	Alleviate bad moods and bring happiness	628	0	5	1.368	1.23	□
	Get a sense of accomplishment	628	0	5	1.42	1.282	□
	Satisfaction	628	0	5	1.393	1.274	□
	Gain recognition and affirmation from others	628	0	5	1.409	1.315	□
	I am interested in participating in time banking and would like to try to experience time banking	628	0	5	1.373	1.246	□
	Realize self-worth	628	0	5	1.377	1.27	□

C. Data Analysis

Before studying the factors that affect Chinese college students' willingness to participate in volunteering in the time bank model, the type of college students' willingness to participate in volunteering should be considered. The survey demonstrates that participants are generally willing to participate in community assistance services and school enrollment promotion activities in exchange for time currency. Compared with volunteer activities for the elderly, respondents considered large event and conference services, environmental protection activities, professional-related volunteer activities, and charity fundraising activities as more worthy of their participation. From the results of the analysis, when formulating the services provided by the college students' time bank, these should not be limited to the existing elderly volunteer service model. Rather, the focus should be on community assistance services and school enrollment publicity activities, which are more likely to enhance the willingness of college students to participate in the time bank.

Second, whether the demand and supply of time banking services match is also worth paying attention to. Most of the participants are willing to exchange their time for credits, and 38.4% of the respondents are willing to exchange their time for career development opportunities, including internship promotion, recommendation letters, and career planning consultation. However, considering that career development opportunities have relatively high requirements on the ability of the participants themselves and the value of opportunities is difficult to measure in time currency, the conversion of time currency to career development opportunities remains challenging

IV. ANALYSIS OF EMPIRICAL RESULTS

A. Reliability Test

Table III. RELIABILITY TEST TABLE

Number of items	Sample size	Cronbach α coefficient
61	628	0.970

Cronbach coefficients are commonly used statistical tools for measuring whether a set of questions (such as a questionnaire) can reliably measure an underlying concept. This questionnaire contained 61 items with 628 sample sizes, and its Cronbach α coefficient is 0.97, indicating that the internal consistency of the questionnaire is very high, and the correlation between items is strong. The overall reliability of the questionnaire is very high.

B. Validity Test

Table IV. KMO Value And Bartlett Sphericity Test

		Overall	Individual level	Family level	Social level	Service and technology
KMO		0.982	0.97	0.5	0.96	0.918
Bartlett Sphericity Test	Approximate Chi-square	30625	5344	328.94	4164	2361.5
	df	1830	55	1	36	15
	p	0	0	0	0	0

The validity analysis results of the questionnaire on Chinese college students' willingness to volunteer in the time bank mode are listed in Table IV. It includes the results of Kaiser–Meyer–Olkin (KMO) sample appropriateness and Bartlett spherical test, which are statistical indicators for evaluating whether the data are suitable for factor analysis.

The overall KMO value of the questionnaire was 0.982, and the KMO values of the individual, social, service and technical levels were all higher than 0.9. This indicates that the commonality between the variables was high, and the factor analysis was appropriate. Because there were only two variables in the family level, namely "My family/friends/classmates want me to participate in the time bank volunteer service" and "My family often volunteers and I want to go," when there are only two variables, the KMO value can only be 0.5.

The Bartlett sphericity test was used to check whether the correlation matrix of the data set is the identity matrix. As listed in the Table IV, at very high confidence ($p = 0$), the correlation matrix of the data set is not the identity matrix, and a certain correlation exists between the variables, which is suitable for factor analysis.

The validity analysis results reveal that the data set of the questionnaire on Chinese college students' willingness to volunteer in the time bank is statistically suitable for factor analysis, which supports the rationality of the questionnaire design and the feasibility of exploring the potential factors (personal, family, social, and service and technology levels). Therefore, the structural validity of the questionnaire in measuring the objective dimension can be deemed good, and it is suitable for further data analysis and interpretation to elucidate the motivation and willingness of Chinese college students to participate in time banking.

V. Correlation Analysis

Table V. Pearson Correlation Analysis

Variables	Willingness to participate in time banking
Gender	0.019
Freshman	0.809**
Student Cadre	0.784**
Political profile	0.682**
Good physical health or not	0.792**
Volunteer experience or not	0.817**
Number of free days a week this semester	0.746**
Volunteering experience is enjoyable or not	0.789**
Willingness to engage in ongoing volunteer service	0.804**
Community Assistance Services	0.069
Supporting education in poverty alleviation	0.036
School enrollment promotion	0.053
Environmental activities	0.013
Professionally related volunteering	0.046
Charity fundraiser	0.039
Volunteering on campus	0.060
Volunteer activities for the aged	0.044
Monthly household income level	0.053
Number of elderly members in the family	0.697**
Knowledge of the time bank	0.777**
Other	0.194**

In this study, the correlation between Chinese college students' willingness to participate in the time bank voluntary service questionnaire was discussed. Through correlation analysis, the mutual relationship and influence degree of each variable were evaluated, as listed in Table V.

The results revealed that the independent variables of the personal, family, society, and service and technology levels all had a significant positive impact on willingness to participate in time banking. In addition, the influence of basic personal information, such as gender, freshman, and political status on the willingness to participate in the time bank was explored.

In terms of personal basic information, gender had no significant impact on college students' willingness to participate in the time bank volunteer service, although previous studies have shown that compared with men, women have stronger empathy and are more willing to participate in volunteer service activities. However, among the surveyed

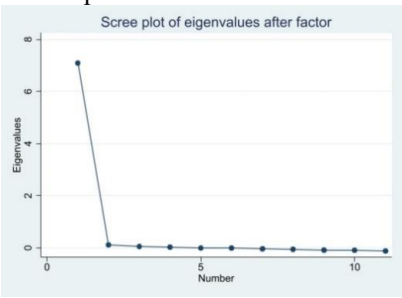
college students, there is no statistically significant difference in the willingness of males and females to participate in time banking. This may be related to the respondents being highly educated and the promotion of the social values of gender equality; further, the decisive factor in college students' willingness to participate in time banking is not empathy but its incentive (such as obtaining credits, commendations, and scholarships). Similarly, the type of volunteer service activity has no significant impact on college students' willingness to participate in the time bank model, which may be because the existing time bank only provides volunteer services for the elderly, and there are no other volunteer services for reference.

On the contrary, the number of elderly members in the nuclear family has a significant negative impact on college students' willingness to participate in the "time bank" volunteer service. Research has demonstrated that the more elderly members in the family, the more willing people are to participate in time banking. However, in this study, the larger the number of elderly members in the family of the Chinese college students surveyed, the more inhibited was their willingness to participate in time banking. Presumably, college students with more elderly members in their family need to spend more time taking care of these family members. This extra responsibility may reduce their time and willingness to participate in other activities, including time banking. College students with more elderly members in their family may have stronger family support networks, which may reduce their reliance on time banking.

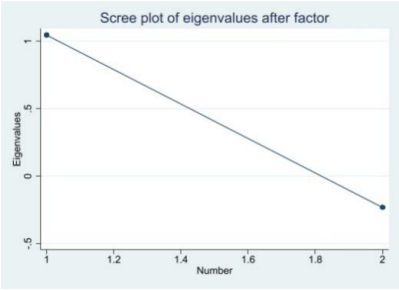
VI. Factor Analysis

The Kaiser criterion is a commonly used criterion in factor analysis, and is centred on the idea of selecting factors with eigenvalues greater than one. This is because, according to this criterion, factors with eigenvalues less than 1 explain less variance than the variance of an original variable, implying that these factors do not contribute enough to explain the data. Therefore, factors with eigenvalues greater than 1 are considered meaningful in that they better represent the structure and variation of the data.

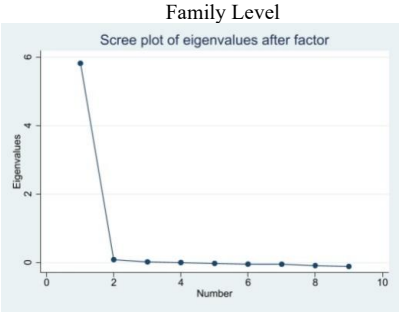
The scree plot in Figure 1 shows a factor with an eigenvalue greater than one at the personal, family, social, and service and technology levels, indicating that these four levels can be respectively condensed into one factor. The maximum variance rotation method (varimax) was used for rotation, and the factor score was predicted.



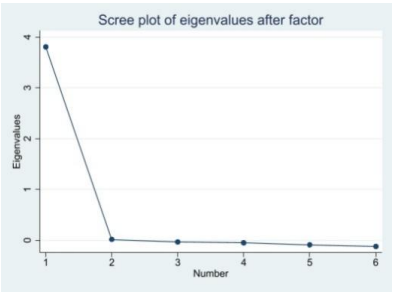
Personal Level



Family Level



Social Level



Service and Technology Level

Figure 1. The Scree Plot at the Personal Level, Family Level, Social Level, Service Level and Technical Level

VII. Multiple Linear Regression Analysis

A multiple linear regression model was established to explore the effects of personal, family, social, and service and technology level factors on Chinese college students' willingness to participate in the time bank voluntary service model. R-squared was 0.738, which suggests that the model has good explanatory force, and all the factors have significantly positive impact on the willingness of volunteers to participate in the time bank model.

Table VI. Multiple Linear Regression Analysis

	(1)
VARIABLES	**Willingness**
p_factor	0.176***
	(0.0323)
f_factor	0.0653***
	(0.0241)
s_factor	0.0700**
	(0.0341)
st_factor	0.116***
	(0.0312)
Constant	0.713***
	(0.00927)
Observations	628
R-squared	0.738
	Standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

However, multicollinearity analysis revealed that the social, personal, service and technology level factor VIF were all up to 10 according to Table VII. Multicollinearity problems were present among variables, which means the regression model was not suitable for independently estimating the impact of each variable on the dependent variable.

Table VII. ANALYSIS OF VARIANCE INFLATION FACTORS

Variable	VIF	1/VIF
s_factor	12.66	0.07901
p_factor	11.5	0.086924
st_factor	10.19	0.098126
f_factor	4.33	0.230858
Mean VIF	9.67	

VIII. RIDGE REGRESSION ANALYSIS

This study utilized ridge regression to solve the collinearity problem of the model. Ridge regression makes the regression coefficient estimable by introducing a k identity matrix. However, introducing an identity matrix leads to information loss, although it can yield a reasonable estimation of the regression model. The smaller the k value is, the smaller the deviation. Combined with the condition of maximum VIF<10 and the ridge trace map, the optimal k value is obtained as 0.01.

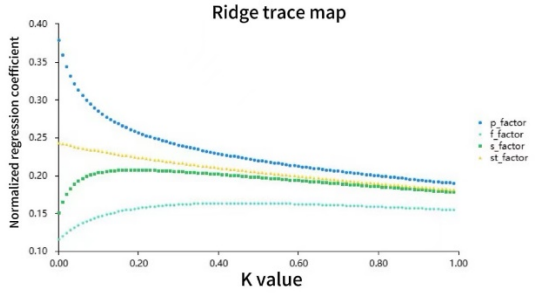


Figure 2. Ridge Trace Map

The ridge regression results are listed in Table VIII, as obtained from the ridge regression model：

The willingness of Chinese college students to participate in time banking = 0.713 + 0.166*personal level factor* + 0.068*family level factor* + 0.077*social level factor* + 0.115*service and technology level factor.*

Notably, the personal, family, social, and service and technology levels all had significant positive impacts on the willingness of college students to participate in time banking. VIF was less than 10 for all factors, there was no multicollinearity problem, and the adjusted R^2 was 0.736, indicating good goodness of fit of the model.

Table VIII. LINEAR REGRESSION ANALYSIS

	Regression coefficient	VIF value
Constant	0.713** (76.950)	-
p_factor	0.166** (5.989)	8.528
f_factor	0.068** (2.963)	3.893
s_factor	0.077** (2.658)	9.146
st_factor	0.115** (4.209)	7.841
Sample size	628	
R^2	0.738	
Adjust R	0.736	
F value	F (4,623)=439.030,p =0.000	

Note: Dependent variable = willingness

* p<0.05 ** p<0.01 The t value is in parentheses

These findings provide an empirical basis for formulating effective time banking volunteer service policies for college students. Policy-makers should focus on improving college students' performance at the personal, family, social, and service and technology levels. Future research should also further explore how to promote cohesion between college students and their families and between college students and society to expand time banking applications.

IX. BINARY LOGISTIC REGRESSION ANALYSIS

Through binary logistic regression analysis, this study further explored the impact of variables such as gender, political status, good health status,prior volunteer experience[37], number of free days per week this semester, whether the volunteering experience is enjoyable[38], willingness to participate in volunteering on an ongoing basis, monthly household income, number of elderly members in family and knowledge of time banking. The influence of the time bank model on the voluntary service intention of Chinese college students was evaluated. The variance inflation factor of these independent variables is less than 10, and there is no multicollinearity problem between the variables.

Table IX. ANALYSIS OF VARIANCE INFLATION FACTORS

Item	VIF value	Tolerance (Tolerance)
Freshmen	3.100	0.323
Political profile	1.643	0.609
Be in good physical health	3.334	0.300
Volunteer experience or not	3.146	0.318
Number of free days a week this semester	2.566	0.390
Whether the volunteering experience is enjoyable	3.100	0.323
Be willing to participate in volunteering on an ongoing basis	3.440	0.291

Table IX. ANALYSIS OF VARIANCE INFLATION FACTORS

Item	VIF value	Tolerance (Tolerance)
Monthly household income level	1.013	0.988
Number of elderly members in family	2.150	0.465
Knowledge of time banking	3.262	0.307

A binary logit regression model was established, and its analysis results are shown in Table X. R-squared is 0.962, and the model has excellent explanatory power. Freshman, political status, good health, volunteer experience, free days in the semester, pleasant volunteer service experience, and willingness to continue to participate in volunteer service have a significant positive impact on the willingness of Chinese college students to participate in time banking. The willingness to participate in time banking is 53 times higher among freshmen compared to the nonfreshmen. Presumably, freshmen are more willing to devote their time to volunteer service activities because they have more free time in a week during the semester [38].Meanwhile, political status also significantly affects the willingness of Chinese college students to participate in time banking. League members and party members are more active participants in time banking. Good health is the decisive factor promoting college students' willingness to participate in time banking. College students with good health are 166 times more willing to participate in time banking than those without good physical health. Variables related to volunteer service, such as having volunteer experience, pleasant volunteer service experience, and willingness to continuously participate in volunteer service, can promote the willingness of college students to participate in time banking.

However, monthly family income does not affect willingness, which may be because college students, by having higher education have a strong sense of social responsibility. In the social environment of mutual help, monthly family income cannot be used as a basis to judge whether they are willing to participate in time banking. Similarly, an understanding of time banking does not have an impact on willingness, possibly because the current mechanism of time banking is still not widely known, so it is impossible to predict whether college students will participate in time banking because they understand.

Table X. ANALYSIS OF BINARY LOGIT REGRESSION RESULTS

Willingness to participate in time banking	β	Odds ratio	Std. err.	P>z	[95% conf.	interval]
Freshmen	3.98	53.38	1.39	0.00	1.26	6.69
Political profile	4.29	73.29	1.44	0.00	1.48	7.11
In good physical health or not	5.11	166.21	2.32	0.03	0.56	9.67
Have volunteer experience or not	4.73	113.82	1.67	0.01	1.47	8.00
Number of free days a week this semester	2.38	10.80	0.82	0.00	0.76	3.99
Volunteering experience is enjoyable or not	3.86	47.55	1.52	0.01	0.89	6.83
Be willing to participate in volunteering on an ongoing basis	3.29	26.78	1.32	0.01	0.71	5.87
Monthly household income level	0.15	1.16	0.60	0.81	1.04	1.33
Number of immediate elder families	0.20	0.82	0.51	0.70	1.19	0.80
Knowledge of time banking	0.04	0.96	0.62	0.95	1.25	1.18
_cons	16.28	0.00	5.54	0.00	27.14	5.42

The overall effectiveness of the model is analyzed in Table XI. The P-value is less than 0.05, which indicates that the original hypothesis is rejected; that is, the independent variable added in the construction of the model has validity, and the construction of the model is meaningful.

Table XI. TEST RESULTS OF THE LIKELIHOOD RATIO OF BINARY LOGIT REGRESSION MODEL

Models	-2 times logarithmic likelihood value	Chi-square value	df	p	AIC	BIC
Intercept only	752					
Final model	29	724	11	0	53	106

As listed in the Table XII, we set "I have willingess to participate in the time bank" as 1 and otherwise set to 0. When the true value is 0, the prediction accuracy rate is 98.33%. In addition, when the true value is 1, the prediction accuracy is 99.33%. The overall prediction accuracy of the model is 99.04%, and the model fits well, indicating that the model can classify the willingness to participate in the time bank with high accuracy and make predictions.

Table XII. PREDICTION ACCURACY OF BINARY LOGIT REGRESSION MODEL

		Predicted Values		Prediction accuracy	Prediction error rate
		0	1		
True Value	0	177	3	98.33%	1.67%
	1	3	445	99.33%	0.67%
Summary				99.04%	0.96%

The goodness-of-fit of the model was judged using the Hosmer–Lemeshow fit test. The original hypothesis of the HL test is that the fitting value of the model is consistent with the observed value. If the P-value is greater than 0.05, it indicates that the model passes the HL test; otherwise, it indicates that the model does not pass the HL test, and the goodness-of-fit of the model is poor. The original assumption of the model test here is that the fitting value of the model is

consistent with the observed value. The p-value here was greater than 0.05 (Chi = 0.468, p = 1.000>0.05), which indicates that the original hypothesis was accepted; that is, the model passes the HL test and has a good degree of model fitting.

Table XIII.HOSMER-LEMESHOW FIT TEST

X^2	Dof df	p value
0.468	8	1.000

X. DISCUSSION

This study aims to explore the willingness of college students to participate in the time bank model of voluntary service and systematically explore the key factors affecting the willingness to participate.

For the hypothesis 1, in this study, the results of personal, family, social, and service and technology levels have significant positive effects on college students' willingness to participate in time banking are consistent with the results of Wu et al.Therefore, the above four aspects should be paid more attention to when formulating the policy of participation time bank for college students. It is necessary to strengthen the personal drive, the family's perception of participating in time bank, the society's positive view of participating in time bank and the mechanism of time bank itself to improve the willingness of college students to participate in time bank.

For hypothesis 2a, in the binary logit regression analysis, there was no indication that the level of knowledge of time banking was related to participation in time banking.

For hypothesis 2b, good health is the decisive factor in college students' willingness to participate in time banking. College students with good health are 166 times more willing to participate in time banking than those with poor health, which is consistent with the research of predecessors that their health level is positively correlated with their willingness to participate in time banking.

Meanwhile, for hypothesis 2c, political status also significantly affects the willingness of Chinese college students to participate in time banking. League members and party members are more active in participating in time banking, which is consistent with time banking participants also being significantly affected by the variable of political status [40].

For hypothesis 2d, variables related to volunteer service, such as having volunteer experience, pleasant volunteer service experience, and willingness to continuously participate in volunteer service, can increase the willingness of college students to participate in time banking.

For hypothesis 2e and 2f, the willingness of freshmen to participate in the time banking mode was 53 times that of not a first-year student, which is consistent with the research finding that freshmen have more free time in one week in the semester and are more willing to devote their time to volunteer service activities [41].

However, the results draw conclusions that are inconsistent with the hypothesis 3a and 3b.

This study found that the number of immediate elders in the family had a significant negative impact on college students' willingness to participate in the time banking model of volunteer service. Some studies have shown that the more elderly members that are present in the immediate family, the more willing to participate in the time bank. However, in the present study, having an elderly member in the immediate family inhibited the willingness of numerous Chinese college students to participate in time banking. Presumably, such students may need to spend more time taking care of or accompanying their elderly family. This extra responsibility may reduce their time and willingness to participate in other activities, including time banking. Moreover, College students with many elderly family members may have stronger family support networks, which may reduce their reliance on time banking.

While research has shown that women are more empathetic and more willing to participate in volunteer activities than men [42], this study found that gender had no significant impact on college students' willingness to participate in the time bank mode of volunteer service. This may be related to the higher level of education of the respondents, the promotion of gender equality values in the society, and the determining factors that motivate university students to participate in time banking are credits or scholarships rather than empathy.

Similarly, monthly family income did not affect willingness, which may be because college students generally receive higher education and have a strong sense of social responsibility. In a social environment of mutual help, monthly family income cannot be used as a basis to judge whether they are willing to participate in time banking. Similarly, understanding time banking did not have an impact on willingness, possibly because the current mechanism of time banking is still not widely known, so it is impossible to predict whether college students will participate in time banking when they understand it.

For Hypotheses 4 and 5, positive social support and a well-developed time banking model can promote college students' willingness to participate in time banking.

Admittedly, this paper has explored the willingness to participate in time banking, but there are still limitations. The existing service model of timebanking, the elderly service, deviates from the community activities and school recruitment campaigns that college students prefer to participate in. In addition, college students are more interested in gaining career development opportunities from time banking. However, considering that career development opportunities have relatively high requirements on the ability of the participants themselves and the value of opportunities is difficult to measure in time currency, the conversion of time currency to career development opportunities remains challenging.

In the future, we should take a deeper look at how to drive the college student population to participate in time banking under the existing time banking model of senior living services. And what kind of policies can be proposed to

more effectively promote the willingness of the university student population to participate in time banking.

XI. Conclusion and Policy Recommendation

This study systematically analyzes the influence of various factors at the individual, family, social, service, and technology levels on Chinese college students' willingness to participate in the "time bank" mode of voluntary service and considers the influence of deeper factors, such as experience value, on college students' willingness to participate. These findings not only highlight the broad prospect of the time bank model in the field of social service but also provide a new perspective for understanding the deep motivation of college students to participate in the time bank model of volunteer service. Additionally, they provide a basis for policy-makers and managers to formulate more effective incentive mechanisms and participation strategies for developing and optimizing the time bank mechanism in the future. In addition, this paper provides a theoretical basis for improving community services to enhance community cohesion and social integration.

The main conclusions are as follows. First, student cadres, political status, whether the volunteer service experience is pleasant, freshmen, good health, volunteer experience, and the number of free in a week during the semester have significant positive effects on willingness. Second, the number of immediate elderly family members has a significant negative impact on willingness. Third, although gender, monthly family income level, and an understanding of time banking have theoretical influence, they do not show significant influence in this study. Fourth, college students are generally willing to participate in community assistance services and school enrollment promotion activities in exchange for time coins. Compared with volunteering for the elderly, they deem it more worthwhile to participate in large-scale events and conference services, environmental protection activities, professional-related volunteering activities, and charity fundraising activities. Therefore, when formulating the services provided by the time bank for college students, they should not be limited to the existing volunteer service model of a time bank for the elderly, and the willingness of college students to participate in the volunteer service of the time bank model should be maximized.

Finally, most college students were willing to exchange their time for credits, and 38.4% of them were willing to exchange their service time for career development opportunities, including internship promotion, recommendation letters, and career planning consultation. Therefore, when formulating policies to attract college students to volunteer in the time bank model, attention should be paid to the importance of incentive content (such as obtaining credits, recognition, and scholarships).

To more effectively attract college students to participate in the time bank mode of volunteer service, promote the volunteer trend, and promote the healthy development of the community, this paper makes the following policy suggestions:

First, the concept of time bank should be introduced early by making use of freshmen's free time, and the significance and benefits of time banking should be publicized in combination with freshmen education activities. The time bank introduction meeting should be held during the orientation week, among other activities, where existing time bank members can share their experiences and feelings. Full play should be given to the enthusiasm of league members and party members. They should be encouraged to actively participate in and publicize the time bank, thus driving more students to participate through their example.

Secondly, considering students with heavy family responsibilities, more flexible service options should be provided to reduce barriers to participation, and service projects suitable for them should be designed.

Third, more types of volunteer service projects, such as community assistance services, school enrollment promotion activities, environmental protection activities, professional-related activities, and charity fundraising, should be introduced to meet the interests and needs of different students. Moreover, flexible service times should be designed and places to adapt to different students' schedules and personal preferences provided.

Finally, fairness among participants should be ensured, feedback collected regularly from participants, the service content and incentive measures of the time bank constantly adjusted and improved, regular evaluation conducted to ensure the project objectives are met, and necessary adjustments made according to the evaluation results. A decentralized multilateral mutual assistance platform should be conducted to match supply and demand and a diversified reward system designed, such as credits, commendations, scholarships, internship opportunities, and letters of recommendation, so that students can use the accumulated time coins in exchange for a variety of resources beneficial to their personal development.

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